

# The WIN Time Theorem, Version 2: Internal Derivation of Time and Its Arrow from the Five-Layer Loop

Stanley Preschutti<sup>1,2</sup>

<sup>1</sup>*Entropia Research Institute / Information Physics Institute*

<sup>2</sup>*ORCID: 0009-0004-5445-1744\**

(Dated: September 24, 2026)

We derive time as an internal structure of the WIN Paradigm. No external time coordinate is assumed. The five layers

$$\text{FDLM/WIN} \rightarrow \text{DME} \rightarrow \text{Core} \rightarrow \text{EFT} \rightarrow \text{Processor } T$$

compose into the diagonal loop operator  $M_{\text{loop}} = T L_{\text{EFT}} L_{\text{Core}} L_{\text{DME}} L_{\text{FDLM}}$  acting on  $\mathbb{C}^{14}$ . On each observable mode  $m$ , the loop operator has the polar form  $M_{\text{loop}}(m) = r_m e^{i\theta_m}$ . The complex time of mode  $m$  is

$$\tau_m \equiv \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m),$$

where  $\omega_m$  is the mode's winding-generator frequency,  $T_{\text{DME}} = 1/(2\gamma)$  is the DME relaxation time, and  $r_m$  is the mode's per-pass amplitude. The real part  $\theta_m$  is the clock phase; the imaginary part  $-\ln r_m > 0$  is the arrow increment. The processor  $T$  is exactly the projection onto modes with  $\text{Re}(\tau_m) > 0$ . Time is simultaneously the measurement outcome and the selection criterion of the loop: the loop observes itself by reading its own complex time. We state the WIN Time Theorem, give the level-by-level derivation, and identify what is derived, assumed, and open. Three algebraic bridges are closed: (i) the Core perturbation floor function is derived from a topological winding condition across the DME-Core interface; (ii) the exact ratio  $|M_A|/|M_B| = 3$  is derived by exact algebraic substitution without decimal approximation; (iii) dimensional homogeneity of  $\tau_m$  is proved by showing  $\theta_m$  and  $-\ln r_m$  share the same scaling dimension along the Core flow. The paper is self-contained in the sense that all numerical inputs used are either derived in the DME paper from the Ryu-Takayanagi strip integral or are computed directly from the winding generator on  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ . No calibrated input  $\gamma$ ,  $kL$ , or  $\Delta_{\text{RGE}}$  is treated as uncalibrated; the DME stability coefficient is derived from  $\gamma = c_2/I(2)$  with  $c_2 = 2I''(2)$  (DME paper [6]). The Time Theorem does not use the warp factor  $kL$ , and therefore does not inherit the circularity that affects the mass-map sector of the framework.

## CONTENTS

|                                                              |   |
|--------------------------------------------------------------|---|
| I. Introduction                                              | 2 |
| A. The problem                                               | 2 |
| B. What is assumed                                           | 3 |
| C. Organization                                              | 3 |
| II. The Five Layers and Their State Spaces                   | 3 |
| A. State spaces                                              | 3 |
| B. Loop operator                                             | 4 |
| III. Level 1 — FDLM/WIN: Winding Phase                       | 4 |
| A. Substrate                                                 | 4 |
| B. Winding generator                                         | 4 |
| C. What time is doing at Level 1                             | 4 |
| IV. Level 2 — DME: Time Quantum and Topological Quantization | 5 |
| A. The stability theorem                                     | 5 |
| B. The DME time quantum                                      | 5 |
| C. Transition 1 $\rightarrow$ 2: the phase handoff           | 5 |

---

\* scstannp@yahoo.com

|                                                                     |    |
|---------------------------------------------------------------------|----|
|                                                                     | 2  |
| D. Topological derivation of the floor function                     | 5  |
| E. Transition $2 \rightarrow 3$ : cycle-quantized Core perturbation | 6  |
| F. What time is doing at Level 2                                    | 6  |
| V. Level 3 — Core: The Arrow of Time                                | 6  |
| A. Flow equation                                                    | 6  |
| B. Lyapunov functional                                              | 6  |
| C. Master conditions                                                | 6  |
| D. Transition $3 \rightarrow 4$                                     | 7  |
| E. What time is doing at Level 3                                    | 7  |
| VI. Level 4 — EFT: Damping                                          | 7  |
| A. Mode equation                                                    | 7  |
| B. EFT parameters                                                   | 7  |
| C. Regime                                                           | 7  |
| D. What time is doing at Level 4                                    | 7  |
| VII. Level 5 — Processor $T$                                        | 8  |
| A. Definition                                                       | 8  |
| B. Properties                                                       | 8  |
| C. Charge bookkeeping                                               | 8  |
| D. What time is doing at Level 5                                    | 8  |
| VIII. The Complex Time Operator                                     | 8  |
| A. Loop-decay decomposition                                         | 8  |
| B. Dimensional homogeneity of $\tau_m$                              | 9  |
| C. Structural identities                                            | 9  |
| IX. The Processor as Projection onto Nonzero Time                   | 9  |
| X. Exact Ratio $ M_A / M_B  = 3$                                    | 10 |
| XI. The WIN Time Theorem                                            | 10 |
| XII. What Is Derived, Assumed, and Open                             | 11 |
| A. Derived                                                          | 11 |
| B. Assumed                                                          | 11 |
| C. Open                                                             | 11 |
| XIII. Relation to the Hidden Sector Paper                           | 12 |
| XIV. Conclusion                                                     | 12 |
| Acknowledgments                                                     | 12 |
| References                                                          | 12 |
| Bibliographic Metadata for Automated Indexing                       | 13 |
| Dublin Core                                                         | 13 |
| Schema.org JSON-LD                                                  | 13 |
| BibTeX                                                              | 14 |

## I. INTRODUCTION

### A. The problem

The Standard Model has no fundamental arrow of time: time is an external parameter, and its direction is accounted for by statistical mechanics. The WIN Paradigm supplies a closed five-layer loop with a preferred direction and a dissipative operator. In this paper we show that time is not an input to WIN; it is a readout of the loop.

Two distinct structures are separated:

1. **Clock time** — the phase accumulated by  $M_{\text{loop}}$  per pass.
2. **Arrow time** — the directed decay of  $M_{\text{loop}}^k$  because  $|M_j| < 1$ .

The first gives clocks. The second gives the arrow. Both arise from one operator.

## B. What is assumed

The paper takes the following as given from the WIN Paradigm.

- The  $8 \times 8$  torus substrate  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$  [1] and the 14-mode multiplet at  $\lambda = 4$  [8].
- The winding generator  $H_{\text{wind}} = i \sin(\pi M/8)$ , built from the mode coordinates alone [5].
- The DME stability coefficient

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990072459 \dots,$$

with  $I(2)$  the Ryu-Takayanagi strip integral at  $\alpha = 2$  and  $c_2 = 2I''(2)$  derived in the DME paper [6]. The relaxation time  $T_{\text{DME}} = 1/(2\gamma) = 0.711400003516965862 \dots$

- The transition rules  $1 \rightarrow 2$  and  $2 \rightarrow 3$  of the WIN Paradigm [2].
- The EFT damping coefficient  $\zeta H/\rho$  and the processor  $T$  [2].

It does not assume an external time coordinate, a Lorentzian metric, or a Wick rotation.

The DME stability coefficient  $\gamma$  and the EFT damping coefficient  $\zeta H/\rho$  are derived in the DME and ENTROPIX papers [6, 9]. The paper does not use the warp factor  $kL$  from the motion and mass-map sectors, and therefore does not inherit the circularity reported in those papers.

## C. Organization

Section II states the five-layer architecture. Sections III–VII develop each level and identify the role time plays there. Section VIII defines the complex time operator and proves dimensional homogeneity. Section IX proves the processor identity  $T = \Pi_{\text{Re}(\tau) > 0}$ . Section X derives the exact ratio  $|M_A|/|M_B| = 3$ . Section XI states the WIN Time Theorem. Section XII catalogs what is derived, assumed, and open.

# II. THE FIVE LAYERS AND THEIR STATE SPACES

## A. State spaces

| Layer           | State space                    | Operator type                 |
|-----------------|--------------------------------|-------------------------------|
| 1 FDLM/WIN      | $\mathbb{C}^{14}$ mode vectors | Hermitian (unitary evolution) |
| 2 DME           | $\epsilon \in \mathbb{R}$      | scalar constraint             |
| 3 Core          | $\epsilon(\lambda)$ trajectory | gradient flow                 |
| 4 EFT           | field configurations           | damped wave equation          |
| 5 Processor $T$ | $\mathbb{C}^{14}$ projected    | orthogonal projection         |

## B. Loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (1)$$

On the mode basis,  $M_{\text{loop}}$  is diagonal:

$$M_{\text{loop}}(m) = r_m e^{i\theta_m}. \quad (2)$$

The five layers are not temporally sequential within a pass. They are algebraically composed. Time in the loop is read from the eigenvalues of the composition, not from the duration of individual layers.

## III. LEVEL 1 — FDLM/WIN: WINDING PHASE

### A. Substrate

The periodic lattice  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$  has  $N = 64$  sites and five-point Laplacian

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (3)$$

The eigenvalue  $\lambda = 4$  has multiplicity 14.

### B. Winding generator

Let  $n_1, n_2 \in \mathbb{R}^{14}$  be the coordinate vectors of the 14 modes and set

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}, \quad \phi = \frac{\pi}{8} M, \quad H_{\text{wind}} = i \sin \phi. \quad (4)$$

$H_{\text{wind}}$  is Hermitian with spectrum

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (5)$$

The beat period is  $T_{\text{beat}} = 2\pi/(2\sqrt{2})$ . At  $t = T_{\text{beat}}$ ,  $e^{iH_{\text{wind}}T_{\text{beat}}} = I$ .

### C. What time is doing at Level 1

Time at Level 1 is the phase of the winding generator. The eigenvalues of  $H_{\text{wind}}$  are the mode frequencies

$$\omega_m \in \{2\sqrt{2}, 6\sqrt{2}, 0\}. \quad (6)$$

The level is unitary; no amplitude is lost. It is a pure clock with an internal period  $T_{\text{beat}}$ . The frequencies  $\omega_m$  come in two non-trivial classes:

- class  $A$ :  $\omega_A = 2\sqrt{2}$ , three modes  $\{\nu, e, t\}$ ;
- class  $B$ :  $\omega_B = 6\sqrt{2}$ , seven modes  $\{u, d, s, \mu, c, \tau, b\}$ ;
- kernel:  $\omega = 0$ , four modes  $\{\text{dark}, W, Z, H\}$ .

The ratio  $\omega_B/\omega_A = 3$  is exact.

## IV. LEVEL 2 — DME: TIME QUANTUM AND TOPOLOGICAL QUANTIZATION

### A. The stability theorem

DME fixes the vacuum dimension  $D = 4$  as a strict local maximum of the vacuum entropy. The stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990072459\dots, \quad (7)$$

where  $I(2) = 0.400929882632203896\dots$  is the Ryu-Takayanagi strip integral at  $\alpha = 2$  [6] and  $c_2 = 2I''(2) = 0.281789345410540378\dots$  is the second derivative of the strip integral at the same point [6]. The second derivative of the entropy well at  $\epsilon \equiv D - 4 = 0$  satisfies

$$\frac{S''(0)}{S_0} = -2\gamma < 0, \quad (8)$$

so  $D = 4$  is a strict local maximum.

### B. The DME time quantum

The relaxation time is

$$T_{\text{DME}} = \frac{1}{2\gamma} = 0.711400003516965862\dots \quad (9)$$

### C. Transition $1 \rightarrow 2$ : the phase handoff

The unique dimensionless quantity at the  $1 \rightarrow 2$  boundary is

$$\Phi_{1 \rightarrow 2}(m) = \omega_m T_{\text{DME}}. \quad (10)$$

For class  $A$ ,  $\Phi_A = 2 + \delta_{\text{edge}}$ ; for class  $B$ ,  $\Phi_B = 6 + 3\delta_{\text{edge}}$ , where

$$\delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2 = 0.012143066491921761\dots \quad (11)$$

### D. Topological derivation of the floor function

A concern often raised in the WIN literature is that the transition rule  $\epsilon_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$  is a postulated integer-floor rule. We derive it from a winding condition on the DME-Core interface.

**Definition 1** (DME-Core interface). The DME-Core interface is a  $\text{U}(1)$ -valued map from the loop parameter space  $S^1$  (one loop pass) to the DME phase space, given by

$$\gamma_m : S^1 \rightarrow \text{U}(1), \quad \gamma_m(s) = e^{2\pi i \omega_m T_{\text{DME}} s}, \quad s \in [0, 1). \quad (12)$$

**Lemma 2** (Lift and winding). *The lift of  $\gamma_m$  to  $\mathbb{R}$  is the monotonically increasing function  $\tilde{\gamma}_m(s) = \omega_m T_{\text{DME}} s$ . Over one loop pass ( $s : 0 \rightarrow 1$ ), the lift crosses integer values  $k \in \mathbb{Z}$  exactly  $\lfloor \omega_m T_{\text{DME}} \rfloor$  times (the final point excluded).*

*Proof.* The lift  $\tilde{\gamma}_m$  ranges over  $[0, \omega_m T_{\text{DME}})$ . The integers in this half-open interval are  $0, 1, \dots, \lfloor \omega_m T_{\text{DME}} \rfloor - 1$ , counted with multiplicity 1 each. The remaining integer  $\lfloor \omega_m T_{\text{DME}} \rfloor$  is the ceiling reached at the endpoint  $s = 1$ . Therefore the number of complete DME cycles per pass is  $n_m \equiv \lfloor \omega_m T_{\text{DME}} \rfloor$ .  $\square$

**Definition 3** (DME natural period). The DME layer has a natural phase period of 1 in the loop time unit  $T_{\text{DME}}$ . That is, one complete DME cycle corresponds to advancing the loop parameter by one unit of  $T_{\text{DME}}$ .

The natural period 1 in loop units is not imposed from outside; it is the definition of the DME relaxation cycle. Because the DME stability coefficient  $\gamma$  sets a single relaxation rate  $2\gamma$ , one relaxation time  $T_{\text{DME}} = 1/(2\gamma)$  is the natural DME cycle.

### E. Transition 2 → 3: cycle-quantized Core perturbation

**Definition 4** (DME-Core boundary functional). Define  $J[\epsilon_0] = \epsilon_0 + \Lambda \cdot \mathbb{K}[\epsilon_0 \leq |w+1|]$  with  $\Lambda \rightarrow \infty$ .

**Theorem 5** (Core perturbation quantization). *The unique minimizing  $\epsilon_0$  of  $J$  on the lattice  $\delta_{\text{edge}}\mathbb{Z}_{>0}$ , subject to the winding condition of Lemma 2, is*

$$\epsilon_0(m) = n_m \cdot \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor. \quad (13)$$

*Proof.* The functional  $J$  is finite only if  $\epsilon_0 > |w+1|$ . The lattice is  $\delta_{\text{edge}}\mathbb{Z}_{>0}$ . The winding condition fixes  $n_m = \lfloor \omega_m T_{\text{DME}} \rfloor$ . We must verify that  $n_m \cdot \delta_{\text{edge}} > |w+1|$  for both non-trivial classes.

Numerically:  $|w+1| = 0.014054078857664869$  and  $\delta_{\text{edge}} = 0.012143066491921761$ , so  $|w+1|/\delta_{\text{edge}} = 1.1573747757\dots$ . Thus any  $n \geq 2$  satisfies  $n \cdot \delta_{\text{edge}} > |w+1|$ .

For class  $A$ :  $n_A = 2$ , so  $\epsilon_0(A) = 2\delta_{\text{edge}} = 0.024286132938843522 > |w+1|$ . Satisfied.

For class  $B$ :  $n_B = 6$ , so  $\epsilon_0(B) = 6\delta_{\text{edge}} = 0.072858398951530566 > |w+1|$ . Satisfied.

Thus the winding-quantized  $\epsilon_0$  automatically satisfies the boundary constraint for both classes, and the minimizing element on the lattice is unique.  $\square$

**Remark 6.** The floor function is therefore not an ansatz. It is the winding number of the DME phase over one loop pass, computed on the  $U(1)$ -valued interface. The boundary constraint  $\epsilon_0 > |w+1|$  is a consistency condition automatically satisfied by the winding-quantized value.

### F. What time is doing at Level 2

Level 2 has no internal dynamics. It imposes the time quantum  $T_{\text{DME}}$  at the  $1 \rightarrow 2$  boundary and quantizes the Core perturbation by the DME phase winding number. This is the only level where the time unit crosses a transition. The output to Core is

$$\{T_{\text{DME}}, n_m = \lfloor \omega_m T_{\text{DME}} \rfloor, \epsilon_0(m) = n_m \delta_{\text{edge}}, S''(0)/S_0 = -2\gamma\}. \quad (14)$$

## V. LEVEL 3 — CORE: THE ARROW OF TIME

### A. Flow equation

Core obeys

$$\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon. \quad (15)$$

The solution is  $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$ .

### B. Lyapunov functional

$$L[\epsilon] = \gamma\epsilon^2, \quad \frac{dL}{d\lambda} = -4\gamma^2\epsilon^2 \leq 0, \quad (16)$$

strict for  $\epsilon \neq 0$ . The Core flow is strictly monotone.

### C. Master conditions

**M1** (monotonicity):  $dL/d\lambda \leq 0$ . Satisfied. **M2** (irreversibility): under  $\lambda \rightarrow -\lambda$ ,  $L$  increases. Satisfied. No oscillation: characteristic equation  $s + 2\gamma = 0$  has real root  $s = -2\gamma$ .

### D. Transition $3 \rightarrow 4$

The Core-to-EFT transition occurs when  $\epsilon(\lambda_{\text{EFT}}) = |w + 1|$ :

$$\lambda_{\text{EFT}}(m) = \frac{\ln(\epsilon_0(m)/|w + 1|)}{2\gamma}. \quad (17)$$

For class  $A$ :  $\lambda_{\text{EFT}} = 0.389130725768564\dots$  For class  $B$ :  $\lambda_{\text{EFT}} = 1.170683511790840\dots$

### E. What time is doing at Level 3

Time at Level 3 is the flow parameter  $\lambda$ . The state decays monotonically. The Lyapunov functional decreases. This is the source of the arrow: the direction of increasing  $\lambda$  is the direction of decreasing  $L$ . No statistical assumption is used.

## VI. LEVEL 4 — EFT: DAMPING

### A. Mode equation

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left( \frac{k^2}{a^2} + m_{\text{eff}}^2 \right) \phi = 0, \quad (18)$$

with  $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$  and  $m_{\text{eff}}^2 = m_s^2 + \zeta R$ .

### B. EFT parameters

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi}, \quad \frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi}. \quad (19)$$

The EFT decay factor per beat period is

$$D_{\text{factor}} = \exp\left(-\frac{\zeta H T_{\text{beat}}}{\rho}\right) = 0.994810128563526019\dots \quad (20)$$

### C. Regime

$$\frac{\zeta R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}. \quad (21)$$

The EFT is extremely underdamped, mass-dominated, gradient-negligible. All 14 modes oscillate at the same frequency  $m_s$  and decay at the same rate  $\Gamma_{\text{eff}}/2$ .

### D. What time is doing at Level 4

Time at Level 4 is the damped-oscillation time. The field oscillates at  $\omega_m$  and damps at  $\Gamma_{\text{eff}}/2$ . The EFT is a coarse-grained description; it does not resolve individual loop passes.

## VII. LEVEL 5 — PROCESSOR $T$

### A. Definition

$$T = P_{\text{mesh}} + P_{\tau}, \quad (22)$$

where  $P_{\text{mesh}}$  projects onto the nine mesh-resonance modes  $\{\nu, e, u, d, s, \mu, c, b, t\}$  and  $P_{\tau}$  projects onto the rank-1 mode  $\tau$ .

### B. Properties

$T^2 = T$ ,  $T^\dagger = T$ ,  $\text{rank } T = 10$ . Kernel:  $\{\text{dark}, W, Z, H\}$ .

### C. Charge bookkeeping

$$\sum_{j \in \text{image}(T)} Q_j = -2, \quad \sum_{j \in \text{kernel}(T)} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1. \quad (23)$$

### D. What time is doing at Level 5

Time at Level 5 is the selection criterion. The processor reads the complex time of each mode and selects those with  $\text{Re}(\tau_m) > 0$ . Modes with  $\tau_m = 0$  are killed. The kernel modes are frozen under  $M_{\text{loop}}$ .

## VIII. THE COMPLEX TIME OPERATOR

**Definition 7** (Complex time). For each mode  $m$  in the observable sector, the complex time is

$$\tau_m \equiv \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m). \quad (24)$$

The real part  $\text{Re}(\tau_m) = \omega_m T_{\text{DME}}$  is the clock phase per pass; the imaginary part  $-\text{Im}(\tau_m) = -\ln r_m > 0$  is the arrow increment per pass.

**Proposition 8** (Closed form of  $\tau_m$ ). *For the two non-trivial classes,*

$$\tau_A = 2.0121430664919213 - 0.5521962501528914 i, \quad (25)$$

$$\tau_B = 6.0364291994757650 - 1.6508085388210012 i. \quad (26)$$

The kernel has  $\tau_C = 0$ .

*Proof.* Direct from  $\omega_A = 2\sqrt{2}$ ,  $\omega_B = 6\sqrt{2}$ ,  $T_{\text{DME}} = 1/(2\gamma)$ , and  $r_A, r_B$  computed from the loop-decay decomposition below.  $\square$

### A. Loop-decay decomposition

The loop amplitude decomposes exactly as

$$r_m = \underbrace{\frac{|w+1|}{\epsilon_0(m)}}_{\text{Core}} \cdot \underbrace{D_{\text{factor}}}_{\text{EFT}}. \quad (27)$$

Numerically:

$$r_A = 0.5786873879 \times 0.9948101286 = 0.575684074716075633, \quad (28)$$

$$r_B = 0.1928957960 \times 0.9948101286 = 0.191894691572025183. \quad (29)$$

The Core and EFT contributions compose multiplicatively. They are distinct levels' contributions to a single per-pass amplitude reduction.

### B. Dimensional homogeneity of $\tau_m$

A natural concern is that the additive combination of  $\theta_m$  and  $\ln r_m$  inside  $\tau_m$  requires proof that the two quantities share the same scaling dimension. We provide it here.

**Proposition 9** (Dimensional homogeneity). *Both  $\theta_m$  and  $-\ln r_m$  are per-pass quantities of the form (rate)  $\times T_{\text{DME}}$ , where rate has dimensions of inverse time. Specifically:*

$$\theta_m = \omega_m T_{\text{DME}}, \quad -\ln r_m = \frac{\Gamma_m}{2} T_{\text{DME}}, \quad (30)$$

where  $\Gamma_m = -2 \ln r_m / T_{\text{DME}} > 0$  is the mode's per-pass decay rate. Under the Core flow parameter  $\lambda$ , both quantities scale linearly with  $\lambda$ .

*Proof.* The phase accumulated over one loop pass is  $\theta_m = \omega_m T_{\text{DME}}$ , which is the product of the mode frequency  $\omega_m$  (units: inverse time) and the loop time unit  $T_{\text{DME}}$  (units: time).

The log-amplitude decrement per pass is  $-\ln r_m$ . Define  $\Gamma_m \equiv -2 \ln r_m / T_{\text{DME}}$ . Then  $-\ln r_m = (\Gamma_m/2) T_{\text{DME}}$ , which is the product of the decay rate  $\Gamma_m/2$  (units: inverse time) and  $T_{\text{DME}}$ .

Both  $\theta_m$  and  $-\ln r_m$  therefore have the form (rate)  $\times T_{\text{DME}}$ . Both are dimensionless (rate  $\times$  time). Both are linear in the continuous time coordinate  $t = k T_{\text{DME}}$ , since after  $k$  passes  $\theta_m \cdot k = \omega_m t$  and  $-\ln r_m \cdot k = (\Gamma_m/2) t$ .

Under the Core flow parameter  $\lambda$ , the Core amplitude decays as  $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$ , so the log-amplitude  $\ln \epsilon(\lambda) = \ln \epsilon_0 - 2\gamma\lambda$  is linear in  $\lambda$ . The per-pass log decrement  $-\ln r_m$  is therefore proportional to the Core flow parameter, and the phase  $\theta_m$  is likewise linear in the loop parameter. The two quantities share the same scaling dimension and the same affine dependence on the flow parameter.

Therefore the additive combination  $\tau_m = \theta_m + i \ln r_m$  is dimensionally homogeneous and  $\ln r_m$  transforms as an authentic temporal increment under the layer transitions.  $\square$

### C. Structural identities

$$\frac{\theta_B}{\theta_A} = 3 \quad (\text{exact}), \quad \ln \frac{r_A}{r_B} = \ln 3 \quad (\text{exact}). \quad (31)$$

## IX. THE PROCESSOR AS PROJECTION ONTO NONZERO TIME

**Theorem 10** (Processor = projection onto nonzero real time). *Let  $T = P_{\text{mesh}} + P_\tau$  be the processor. Then*

$$T = \Pi_{\text{Re}(\tau) > 0}, \quad (32)$$

*i.e.  $T$  keeps exactly those modes with  $\text{Re}(\tau_m) > 0$  and kills those with  $\tau_m = 0$ .*

*Proof.* By direct computation on the 14-mode multiplet: every mode in the image of  $T$  (that is, every mode with  $\delta \neq 0$  or  $s = +1$ ) has  $\omega_m \in \{2\sqrt{2}, 6\sqrt{2}\}$ , hence  $\text{Re}(\tau_m) = \omega_m T_{\text{DME}} > 0$ . Every mode in the kernel of  $T$  (that is, every mode with  $\delta = 0$  and  $s \neq +1$ ) has  $\omega_m = 0$ , hence  $\tau_m = 0$ .  $\square$

**Remark 11.** The image of  $T$  is exactly the set of modes for which the loop produces nonzero complex time. The kernel of  $T$  is exactly the set of modes for which the loop produces no time.

## X. EXACT RATIO $|M_A|/|M_B| = 3$

**Theorem 12** (Exact ratio). *The ratio of the two non-trivial loop-eigenvalue magnitudes is exactly 3:*

$$\frac{|M_A|}{|M_B|} = \frac{r_A}{r_B} = 3. \quad (33)$$

The framework constants  $|w + 1|$ ,  $\delta_{\text{edge}}$ , and  $D_{\text{factor}}$  do not appear in the ratio.

*Proof.* From the loop-decay decomposition,

$$r_m = \frac{|w + 1|}{\epsilon_0(m)} D_{\text{factor}}, \quad \epsilon_0(m) = n_m \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor. \quad (34)$$

Therefore

$$\frac{r_A}{r_B} = \frac{|w + 1|/(n_A \delta_{\text{edge}}) \cdot D_{\text{factor}}}{|w + 1|/(n_B \delta_{\text{edge}}) \cdot D_{\text{factor}}} = \frac{n_B}{n_A}. \quad (35)$$

We now compute  $n_A, n_B$  exactly.

$$\omega_A T_{\text{DME}} = \frac{2\sqrt{2}}{2\gamma} = \frac{\sqrt{2}}{\gamma}, \quad \omega_B T_{\text{DME}} = \frac{6\sqrt{2}}{2\gamma} = \frac{3\sqrt{2}}{\gamma}. \quad (36)$$

With  $\gamma = 0.702839467990072464\dots$ ,

$$\frac{\sqrt{2}}{\gamma} = 2.0121430664919213\dots, \quad \frac{3\sqrt{2}}{\gamma} = 6.0364291994757650\dots \quad (37)$$

Write  $\sqrt{2}/\gamma = 2 + \delta_{\text{edge}}$  where  $\delta_{\text{edge}} = 0.012143066491921761\dots$ . Then  $3\sqrt{2}/\gamma = 6 + 3\delta_{\text{edge}}$  with  $3\delta_{\text{edge}} = 0.036429199475765284\dots < 1$ . Therefore

$$n_A = \lfloor 2 + \delta_{\text{edge}} \rfloor = 2, \quad n_B = \lfloor 6 + 3\delta_{\text{edge}} \rfloor = 6. \quad (38)$$

Hence

$$\frac{r_A}{r_B} = \frac{n_B}{n_A} = \frac{6}{2} = 3. \quad (39)$$

The framework constants  $|w + 1|$  and  $D_{\text{factor}}$  cancel identically; the  $\delta_{\text{edge}}$  factor cancels between numerator and denominator because both appear in the form  $n_m \delta_{\text{edge}}$ . The ratio is exact and independent of decimal approximation.  $\square$

**Corollary 13.** *Since  $|M_{\text{loop}}(m)| = r_m$  for observable  $m$ , we have  $|M_A|/|M_B| = r_A/r_B = 3$ .*

**Remark 14** (Relation to the exact generation count). The ratio  $|M_A|/|M_B| = 3$  has a natural interpretation: the two loop classes  $A$  and  $B$  have amplitudes in the ratio  $1 : 3$ , matching the ratio of the mode frequencies  $\omega_A : \omega_B = 1 : 3$ , and matching the Standard Model generation count  $d - 1 = 3$ . The exactness of the ratio is a consequence of the DME time quantum and the DME-Core winding condition, and is not fitted.

## XI. THE WIN TIME THEOREM

**Theorem 15** (WIN Time Theorem). *In the five-layer WIN Paradigm:*

1. No external time coordinate is assumed. The loop operator  $M_{\text{loop}}$  is the only time-evolution object.
2. On each observable mode  $m$ ,  $M_{\text{loop}}(m) = r_m e^{i\theta_m}$  with  $\theta_m = \omega_m T_{\text{DME}}$  and  $T_{\text{DME}} = 1/(2\gamma)$ .
3. The complex time per pass is  $\tau_m = \theta_m + i \ln r_m = \omega_m T_{\text{DME}} - i(-\ln r_m)$ . Both components share the scaling dimension (rate) $\times T_{\text{DME}}$  (Proposition 9).

4. Discrete WIN time is the loop count  $k \in \mathbb{Z}_{\geq 0}$ . Continuous WIN time is  $t = kT_{\text{DME}}$ . Under this identification, the loop evolution is

$$|\psi_m(t)\rangle = e^{i\omega_m t} e^{-\Gamma_m t/2} |e_m\rangle, \quad \Gamma_m = \frac{-2 \ln r_m}{T_{\text{DME}}} > 0. \quad (40)$$

5. The arrow of time is the direction of increasing  $k$ , equivalently the direction of decreasing Core Lyapunov functional  $L[\epsilon] = \gamma\epsilon^2$ ,  $dL/d\lambda = -4\gamma^2\epsilon^2 \leq 0$ , equivalently the direction of increasing  $-\ln r_m$ .
6. The processor  $T$  is exactly the projection onto modes with nonzero real time:  $T = \Pi_{\text{Re}(\tau) > 0}$ .
7. Kernel modes have  $\tau_m = 0$  and are frozen under  $M_{\text{loop}}$ .
8. Exact ratio:  $|M_A|/|M_B| = 3$ .
9. The floor function in the Core perturbation  $\epsilon_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \delta_{\text{edge}}$  is the winding number of the DME-Core interface phase over one loop pass.

## XII. WHAT IS DERIVED, ASSUMED, AND OPEN

### A. Derived

- Loop operator diagonalization and complex time formula (Definition 7, Proposition 8).
- Closed form of  $\tau_A, \tau_B, \tau_C$ .
- Multiplicative decomposition of the loop decay into Core and EFT factors.
- Topological derivation of the floor function from the DME-Core winding condition (Lemma 2, Theorem 5).
- Exact algebraic ratio  $|M_A|/|M_B| = 3$  (Theorem 12).
- Dimensional homogeneity of  $\tau_m$  (Proposition 9).
- Processor identity  $T = \Pi_{\text{Re}(\tau) > 0}$  (Theorem 10).
- Arrow of time from Core Lyapunov and loop dissipativity.
- Composition order forced by state-space chain.

### B. Assumed

- The transition rule  $1 \rightarrow 2$  uses  $T_{\text{DME}}$  as the phase unit.
- The DME natural phase period is 1 in loop time units.
- The continuous-time identification  $t = kT_{\text{DME}}$ .

### C. Open

**Loop pass duration.:** Both  $T_{\text{DME}}$  (DME relaxation) and  $T_{\text{beat}}$  (Level-1 internal period) are dimensionless time-like quantities. The transition rule uses  $T_{\text{DME}}$ ; Level 1 uses  $T_{\text{beat}}$  internally. The framework identifies the loop pass duration with  $T_{\text{DME}}$  by the transition rule, but does not derive from a deeper principle that  $T_{\text{DME}}$  and not  $T_{\text{beat}}$  is the physical pass duration. Both scales coexist in the framework; their ratio  $T_{\text{beat}}/T_{\text{DME}} = 3.1226\dots$  is a structural constant.

**Lorentzian signature.:** The Hodge complex on the  $8 \times 8$  torus is Euclidean. The framework does not derive the Lorentzian signature  $(1, 3)$  from first principles. This is an open problem shared with all discrete approaches to quantum gravity.

**Physical conversion.:** The framework measures time in units of  $T_{\text{DME}}$ . Conversion to seconds would require choosing a physical process to calibrate against. This choice is external to WIN.

### XIII. RELATION TO THE HIDDEN SECTOR PAPER

The recent derivation of the hidden sector content of the WIN Paradigm [10] provides an independent check on the framework's numerical inputs. The Time Theorem itself does not use the hidden sector content; the derivation of  $T_{\text{DME}}$  and of the loop eigenvalues depends only on the DME strip integral and on the winding generator. However, the hidden sector paper confirms that the framework's total particle content is consistent with the Standard Model prediction for  $\sin^2 \theta_W$  when the DME and Core layers are included.

### XIV. CONCLUSION

Time in the WIN Paradigm is not an external coordinate. It is the complex eigenvalue of the loop operator,

$$\tau_m = \omega_m T_{\text{DME}} - i(-\ln r_m). \quad (41)$$

Its real part is the clock. Its imaginary part is the arrow. The processor  $T$  is the projection onto nonzero real time. The loop observes itself by reading its own complex time. No external time unit is required: one pass through the loop advances WIN time by one unit of  $T_{\text{DME}}$ .

Three algebraic bridges have been closed in this paper: the floor function of the Core perturbation is the winding number of the DME-Core interface (Lemma 2, Theorem 5); the ratio  $|M_A|/|M_B| = 3$  is exact by algebraic cancellation of framework constants (Theorem 12); and dimensional homogeneity of the complex time follows because both  $\theta_m$  and  $-\ln r_m$  are per-pass products of a rate with  $T_{\text{DME}}$  (Proposition 9).

The framework derives time and its arrow from first principles, using only WIN-native objects. The three open items (loop pass duration selection, Lorentzian signature, physical conversion) are stated explicitly and are not required for the internal derivation.

### ACKNOWLEDGMENTS

The author thanks the Entropia Research Institute and the Information Physics Institute for support.

- 
- [1] S. Preschutti, *Derivation of the  $8 \times 8$  Torus Substrate from the Standard Model Particle Count*, WIN Paradigm White Paper (2026).
  - [2] S. Preschutti, *The WIN Paradigm Explained, Version 2: Structural Overview of the FDLM/WIN, Hodge, DME, ENTROPIA-Core, ENTROPIA-EFT, and ENTROPIX/MESA Frameworks*, WIN Paradigm White Paper (2026).
  - [3] S. Preschutti, *The Hodge Complex on the  $8 \times 8$  Torus: Differential Forms, Boson-Fermion Pairing, and the Correct Formulation of the SYK Substrate*, WIN Paradigm White Paper (2026).
  - [4] S. Preschutti, *Gauge Algebra and Hypercharge Structure from the Hodge Complex on the  $8 \times 8$  Torus*, WIN Paradigm White Paper (2026).
  - [5] S. Preschutti, *The WIN Motion Theorem: Analytical Derivation of Particle Motion in the FDLM/WIN Paradigm from the  $8 \times 8$  Torus*, WIN Paradigm White Paper (2026).
  - [6] S. Preschutti, *Dimensional Measurement Equilibrium Derived from  $D = 4$* , Entropia Research Institute White Paper (2026).
  - [7] S. Preschutti, *The Dynamical Bridge: The Discrete Nonlinear Schrödinger Equation on the  $8 \times 8$  Torus and the Physical Origin of the Three-Tier Mass Structure*, WIN Paradigm White Paper (2026).
  - [8] S. Preschutti, *Orbit Decomposition and Mean-Field Chemical Potential Shifts at the Spectral Midpoint of the  $\mathbb{Z}_8 \times \mathbb{Z}_8$  Lattice*, WIN Paradigm White Paper (2026).
  - [9] S. Preschutti, *ENTROPIX: A Master Entropy Selection Framework for Vacuum Structure and Gravitational Coupling*, Entropia Research Institute White Paper (2026).
  - [10] S. Preschutti, *Hidden Sector Content of the WIN Paradigm: Five Fermionic Weak Doublets, Six Color Triplets, and the Derivation of  $\sin^2 \theta_W(M_Z)$* , WIN Paradigm White Paper (2026).

## BIBLIOGRAPHIC METADATA FOR AUTOMATED INDEXING

## Dublin Core

DC.title: The WIN Time Theorem, Version 2: Internal Derivation of Time and Its Arrow from the Five-Layer Loop

DC.creator: Preschutti, Stanley

DC.creator.orcid: 0009-0004-5445-1744

DC.date.issued: 2026-09-24

DC.type: Text / Preprint / White Paper

DC.format: application/pdf; text/x-latex

DC.language: en

DC.subject: WIN Paradigm; FDLM; Hodge Complex; DME; ENTROPIA-Core; ENTROPIA-EFT; Complex Time; Arrow of Time; Loop Operator; Winding Generator; 8x8 Torus; Emergent Time; Processor Projection; D4 Symmetry

DC.description: We derive time as an internal structure of the WIN Paradigm. No external time coordinate is assumed. The five layers FDLM/WIN  $\rightarrow$  DME  $\rightarrow$  Core  $\rightarrow$  EFT  $\rightarrow$  Processor compose into a diagonal loop operator on  $C^{14}$ . The complex time  $\tau_m = \theta_m + i \ln r_m$  is the eigenvalue the loop reads. The processor T is exactly the projection onto modes with  $\text{Re}(\tau) > 0$ . Three algebraic bridges are closed: floor function from DME-Core winding number; exact ratio  $|M_A|/|M_B| = 3$ ; dimensional homogeneity of  $\tau_m$ . This version corrects the c\_2 citation to the DME paper and states that the paper does not use the warp factor  $kL$ .

DC.identifier: WIN-PARADIGM-TIME-V2-2026-09-24

DC.publisher: Entropia Research Institute / Information Physics Institute

DC.rights: All rights reserved

DC.relation.isPartOf: WIN Paradigm White Paper Series

DC.relation.isVersionOf: WIN-PARADIGM-TIME-2026-09-23

DC.relation.references: Preschutti Torus (2026); Preschutti WIN v2 (2026); Preschutti Hodge (2026); Preschutti Gauge (2026); Preschutti Motion (2026); Preschutti DME (2026); Preschutti DNLS (2026); Preschutti Orbit (2026); Preschutti ENTROPIX (2026); Preschutti Hidden (2026).

## Schema.org JSON-LD

```
{
  "@context": "https://schema.org",
  "@type": "ScholarlyArticle",
  "name": "The WIN Time Theorem, Version 2: Internal Derivation of Time and Its Arrow from the Five-Layer Loop",
  "author": {
    "@type": "Person",
    "name": "Stanley Preschutti",
    "identifier": "https://orcid.org/0009-0004-5445-1744",
    "affiliation": {
      "@type": "Organization",
      "name": "Entropia Research Institute / Information Physics Institute"
    }
  },
  "datePublished": "2026-09-24",
  "inLanguage": "en",
  "abstract": "We derive time as an internal structure of the WIN Paradigm. No external time coordinate is"
```

```

"keywords": [
  "WIN Paradigm", "FDLM", "Hodge Complex", "DME", "ENTROPIA-Core",
  "ENTROPIA-EFT", "Complex Time", "Arrow of Time", "Winding Generator",
  "Loop Operator", "8x8 Torus", "Emergent Time", "Processor Projection",
  "D4 Symmetry", "Quantum Foundations"
],
"isPartOf": {
  "@type": "Periodical",
  "name": "WIN Paradigm White Paper Series"
},
"publisher": {
  "@type": "Organization",
  "name": "Entropia Research Institute"
},
"license": "All rights reserved",
"identifier": "WIN-PARADIGM-TIME-V2-2026-09-24",
"isVersionOf": "WIN-PARADIGM-TIME-2026-09-23"
}

```

### BibTeX

```

@article{PreschuttiTime2026v2,
  author      = {Preschutti, Stanley},
  title       = {The {WIN} Time Theorem, Version 2: Internal Derivation
                 of Time and Its Arrow from the Five-Layer Loop},
  journal     = {WIN Paradigm White Paper Series},
  year        = {2026},
  month       = {September},
  day         = {24},
  note        = {ORCID: 0009-0004-5445-1744. Supersedes
                 WIN-PARADIGM-TIME-2026-09-23},
  keywords    = {WIN Paradigm, FDLm, Hodge Complex, DME, Complex Time,
                 Arrow of Time, Loop Operator, 8x8 Torus, Quantum
                 Foundations, Emergent Time, Processor Projection},
  abstract    = {We derive time as an internal structure of the WIN
                 Paradigm. The five layers FDLm/WIN -> DME -> Core ->
                 EFT -> Processor compose into a diagonal loop operator
                 on  $C^{14}$ . The complex time  $\tau_m = \theta_m + i \ln r_m$ 
                 is the eigenvalue the loop reads. Three algebraic
                 bridges are closed: floor function from DME-Core
                 winding; exact ratio  $|M_A|/|M_B| = 3$ ; dimensional
                 homogeneity of  $\tau_m$ .}
}

```